Epilepsy & Behavior 57 (2016) 82–89

Contents lists available at ScienceDirect

Epilepsy & Behavior

journal homepage: www.elsevier.com/locate/yebeh

Review

Non-EEG based ambulatory seizure detection designed for home use: What is available and how will it influence epilepsy care?

Judith van Andel [a,*], Roland D. Thijs [b,c], Al de Weerd [d], Johan Arends [e], Frans Leijten [a]

[a] University Medical Centre Utrecht, Department of Clinical Neurophysiology, Utrecht, The Netherlands
[b] Stichting Epilepsie Instellingen Nederland SEIN, Department of Clinical Neurophysiology, Heemstede, The Netherlands
[c] Leiden University Medical Centre, Department of Neurology, Leiden, The Netherlands
[d] Stichting Epilepsie Instellingen Nederland SEIN, Department of Clinical Neurophysiology, Zwolle, The Netherlands
[e] Academic Centre for Epileptology Kempenhaeghe, Department of Clinical Neurophysiology, Heeze, The Netherlands

ARTICLE INFO

Article history:
Received 15 June 2015
Revised 31 December 2015
Accepted 2 January 2016
Available online 27 February 2016

Keywords:
Seizure detection

ABSTRACT

Objective: This study aimed to (1) evaluate available systems and algorithms for ambulatory automatic seizure detection and (2) discuss benefits and disadvantages of seizure detection in epilepsy care.
Methods: PubMed and EMBASE were searched up to November 2014, using variations and synonyms of search terms "*seizure prediction*" OR "*seizure detection*" OR "*seizures*" AND "*alarm*".
Results: Seventeen studies evaluated performance of devices and algorithms to detect seizures in a clinical setting. Algorithms detecting generalized tonic–clonic seizures (GTCSs) had varying sensitivities (11% to 100%) and false alarm rates (0.2–4/24 h). For other seizure types, detection rates were low, or devices produced many false alarms. Five studies externally validated the performance of four different devices for the detection of GTCSs. Two devices were promising in both children and adults: a mattress-based nocturnal seizure detector (sensitivity: 84.6% and 100%; false alarm rate: not reported) and a wrist-based detector (sensitivity: 89.7%; false alarm rate: 0.2/24 h).
Significance: Detection of seizure types other than GTCSs is currently unreliable. Two detection devices for GTCSs provided promising results when externally validated in a clinical setting. However, these devices need to be evaluated in the home setting in order to establish their true value. Automatic seizure detection may help prevent sudden unexpected death in epilepsy or status epilepticus, provided the alarm is followed by an effective intervention. Accurate seizure detection may improve the quality of life (QoL) of subjects and caregivers by decreasing burden of seizure monitoring and may facilitate diagnostic monitoring in the home setting. Possible risks are occurrence of alarm fatigue and invasion of privacy. Moreover, an unexpectedly high seizure frequency might be detected for which there are no treatment options. We propose that future studies monitor benefits and disadvantages of seizure detection systems with particular emphasis on QoL, comfort, and privacy of subjects and impact of false alarms.

1. Introduction

Despite the rapid expansion of pharmaceutical and surgical treatment options, a significant proportion of people with epilepsy continue to have seizures. They are faced with the prospect of having to cope with seizures for a long time, if not lifelong. Forty years ago, the prevalence of refractory epilepsy was estimated at approximately 30% [1], and this percentage has not changed substantially since then [2]. The mortality rate, which is 2–4 times higher in people with epilepsy than in their peers without epilepsy, has also not changed over the last decades [3–5]. Epilepsy adversely affects quality of life (QoL), with unsuccessful achievement of seizure freedom, passive coping styles, and low socioeconomic status being major determinants of an impaired QoL [6]. In our experience, many people struggle with the random nature of seizures. Moreover, seizures are dangerous, potentially leading to falls, violent movements, confusional wandering, sudden unexpected death in epilepsy (SUDEP), or status epilepticus.

There is an urgent need for alternative strategies to reduce the burden of epilepsy. Devices to detect seizures have been promoted as a way to reduce seizure-related risk, but the scientific validity of this notion is questionable, and the implications for epilepsy care have not been investigated. Here, we describe various in-home seizure detection devices and present the results of a literature search of current systems for automatic detection. We then discuss the risks and benefits of continuous automated seizure monitoring and describe the ideal setup of future devices for seizure detection.

* Corresponding author. Tel.: +31 88 755 6782.
E-mail address: judithvanandel@gmail.com (J. van Andel).

http://dx.doi.org/10.1016/j.yebeh.2016.01.003
1525-5050/© 2016 Published by Elsevier Inc.

2. Devices for automatic seizure detection

2.1. EEG

Many research groups have worked on the development of EEG algorithms to automatically predict seizures or detect seizure onset in people with epilepsy. Seizure prediction aims to rapidly detect early seizure activity, with a view to instigating timely interventions to prevent further seizure activity. Automatic seizure prediction has proven difficult [7]. A recent long-term ambulatory study of intracranial EEG recordings [8] yielded promising results regarding prediction accuracy in specific subjects, but there is still a long way to go before this method can be used clinically, mainly because interventional techniques, such as closed-loop stimulation and fast-working locally administered medication, are still under investigation. The method is invasive and costly, requiring implantation of an intracranial device, accurate localization of the seizure-onset zone, and definition of the EEG dynamics of seizure onset. For these reasons, it will be reserved for a small number of people with refractory epilepsy.

Automatic seizure detection with scalp EEG focuses on a more efficient analysis of long-term, diagnostic EEG data, as collected in presurgical screening of people with epilepsy. Many studies have been performed to develop reliable algorithms. The main issue in development is reducing the number of false positives. In early algorithms, the false positive rates were up to 5 false detections per hour [9]. False positives are mainly caused by artifacts in the EEG signal often because of muscle activity. More recently developed algorithms include methods of artifact rejection, greatly improving false alarm rates. Several algorithms reach sensitivities >80% and false alarm rates <0.5 false alarms/h [10–12]. Artifact rejection could lead to decreased sensitivity in detection of seizures where high amounts movement are present from seizure onset. However, evaluation in long-term data of a recent algorithm presented promising results [10]. Development of new algorithms benefits from modern techniques of classification, such as random forest classifiers or techniques from genome analysis [13,14]. No system based on automatic EEG analysis is currently used in clinical practice. In theory, these algorithms could also be used online in an ambulatory setting. In this context, it is important to consider that real-time classification of seizures with a portable system requires an algorithm with relatively low computational complexity [15]. Also, long-term scalp EEG is uncomfortable, sensitive to artifacts, and cosmetically difficult to hide from the outside world, making its application impractical in daily life. New developments include 4-electrode subcutaneous EEG recording, which is currently being evaluated for the detection of hypoglycemia in diabetes [16]. This review will focus on non-EEG-based systems.

2.2. Non-EEG sensors

As EEG recording is currently too cumbersome for in-home seizure monitoring, systems based on extracerebral signals have been developed, based on seizure-induced movements and autonomic changes [17]. Movement and position can be recorded with accelerometers, gyroscopes (measuring rotational acceleration), or magnetometers (measuring position) attached to the body. Movements caused by nocturnal or sleep-related seizures can also be detected with a mattress-based sensor, which is less bothersome than on-body sensors. Other contact-free technologies are radar and video. With radar signals, the position of the body can be monitored continuously and, therefore, also changes in position due to movement. A video camera can also be used in combination with automatic movement analysis, for instance, using optical flow analysis [18]. Video or radar technologies are contact-free and, therefore, less burdensome, but can only be used in the room in which the equipment is set up. Electromyography is used to measure muscle contractions, which is especially useful in assessing the tonic phase of seizures. Electromyography requires electrodes attached to the skin, which can be sensitive to artifacts or uncomfortable to wear. Electrocardiography (ECG) or plethysmography can be used to measure changes in heart rate and heart rate variability. Two-lead ECG can also be used to detect ECG abnormalities. Electrocardiography also requires electrodes which can be sensitive to artifacts. Another option to assess autonomic changes is through measurement of electrodermal activity. Electrodermal activity (EDA) can be measured by assessing conductance between two electrodes on the skin. It changes as a result of sweating, which is regulated by the sympathetic nervous system. Changes in respiration can also be interesting parameters for seizure detection, especially when the goal is detection of clinically urgent seizures. A pulse oximeter can be used to measure respiration-related changes in blood oxygenation. Respiration is usually measured with body sensors assessing the movement of the chest wall or with a mask or sensor placed over or under the nose to measure airflow. Video and radar can also be used for this purpose, and it even seems possible to measure heart rate with these techniques [19]. These methods have their advantages and disadvantages for seizure detection; for example, changes in sensitivity, false-positive seizure detection due to autonomic changes as a result of arousal or nightmares, the possibility to monitor vital signs, susceptibility to technical failure, difficulty of instruction, energy consumption, and burden and side effects to users. When considering non-EEG sensors for long-term monitoring, it is important to keep these sensors wearable. Currently, there is a surge in wearable sensors in watches, bands, or clothing in a commercial setting. These sensors combine several modalities in one compact sensor which can easily be worn around a wrist or an upper arm. This provides researchers with opportunities for commercial collaboration or to develop integrated sensor systems with existing technology.

2.3. Clinical evaluation of available systems

A number of products are marketed for automatic, non-EEG, in-home seizure detection. Some systems focus on nocturnal seizure detection and other systems on continuous use. Though they are becoming more popular, in practice, these products are still infrequently used because of detection failures, frequent false alarms, and limited scientific proof of efficacy [19].

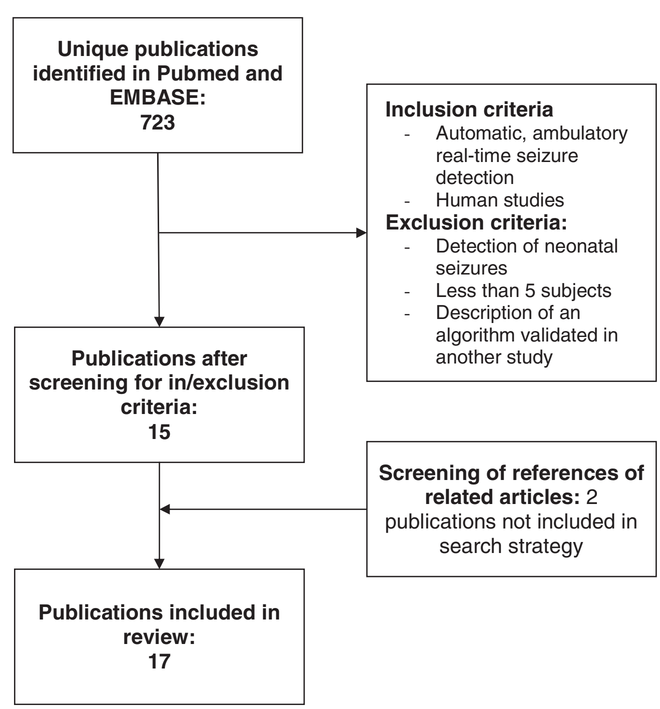

Fig. 1. Flow chart of search strategy.

Table 1
Summary of results of studies reporting performance of automatic, non-EEG, seizure detection algorithms and devices.

	Modality	Setup	N patients (with seizures)	N seizures	Seizure type (N seizures)	Hours of recording	Outcome[a]	Intention to diagnose[b]	External validation	Remark
Nijsen et al. [20]	ACM	Sensor on one arm	36 (18)	27	T	NR	Sens: 80%PPV: 35%	No	No	42% of false alarms were myoclonic or clonic seizures.
Nijsen et al. [21]	ACM	Sensors on arms	36 (36)	64	M	NR	Sens: 80%PPV: 16%	Yes	No	Algorithm trained on 29 seizures and tested on 35 seizures.
Kramer et al. [22]	ACM	1 sensor in bracelet on limb	31 (15)	22	C, GTC (16)	1692	Sens: 91%Latency: 17sFA: 0.1/24h	No	No	Pts with dystonic posturing or behavioral automatisms were excluded; available as Epilert® (BioLert, Ltd., Even Yehuda, Israel).
Lockman et al. [24]	ACM	Sensor wrist/ankle	40 (6)	8	GTC	NR	Sens: 88%	Yes	No	204 FA canceled by pts, 1 nocturnal FA; available as SmartWatch® (SmartMonitor, San Jose, USA).
Beniczky et al. [25]	ACM	Sensor on wrist	73 (20)	39	GTC	4878	Sens: 89.7%FA: 0.2/24hLatency: 55s	Yes	Yes	Available as Epi-Care Free® (Danish Care Technology, Sorø, Denmark).
Van de Vel et al. [23]	ACM	Sensor on extremities	7 (7)	51	HM	NR	Sens: 95.7%PPV: 57.84%	Yes	Yes	System targeted to children. Only nocturnal data included. No patient-specific information necessary for detection algorithm.
Luca et al. [32]	ACM	Sensor on extremities	7 (7)	51	HM	NR	Average sens: 85%Average PPV: 46%	Yes	–	Only nocturnal data included. Unsupervised method based on extreme value statistics. No labeled trianing data necessary.
Poh et al. [26]	ACM, EDA	Wristworn sensor	80 (7)	16	GTC	4213	Sens: 94%Latency: 31.4sFA: 0.74/24h	Yes	No	Algorithm based on patient-specific information, scored baseline signal necessary.
Narechania et al. [34]	Mvt	Pressure sensor in mattress	51 (13)	18	GTC	3741	Sens: 89%PPV: 43%Latency: 9s in clonic phase	No	Yes	Commercially available, Emfit® (Emfit Ltd, Vaajakoski, Finland). Only adult pts. 8 nocturnal GTCSs recorded (sens: 100%, no nocturnal FA).
Van Poppel et al. [28]	Mvt	Pressure sensor in mattress	45 (26)	78	GTC (13)GM, T, CP	NR	Overall sens: 29.5%;Sens GTC: 84.6%FA: NR	Yes	Yes	Commercially available, Emfit® (Emfit Ltd, Vaajakoski, Finland). Only pediatric pts. nocturnal detection.
Fulton et al. ST2® [30]	Mvt	Sensor under mattress	27 (15)	69	GTC (3), GM, T, CP	NR	Overall sens: 2.2%Sens GTC: 0/3	Yes	Yes	ST2® (MedPage, no longer commercially available) was used in unspecified subset of patients. Nocturnal detection.
Carlson et al. [27]	Mvt, sound	Sensor + microphone under mattress	64 (6)	8	GTC	1528	Sens: 62.5% (5/8)FA: 4/24h	Yes	Yes	No individual calibration applied. Available as MP5® (MedPage, Corby, UK). Nocturnal detection.
Fulton et al. MP5® [30]	Mvt, sound	Sensor + microphone under mattress	27 (15)	69	GTC (9), GM, T, CP	NR	Overall sens: 4.3%Sens GTC: 11.1% (1/9)	Yes	Yes	MP5® was used in unspecified subset of patients. 4 patients found device too uncomfortable for long-term use. Nocturnal detection.
Conradsen et al. [31]	EMG	Electrode on left deltoid muscle	60 (11)	22	GTC	776	Sens: 100%Latency: 13.7FA: 1/24h	No	No	In future available as IctalCare® (IctalCare, Hørsholm, Denmark).
Van Elmpt et al. [33]	Heart rate	ECG	17 (16)	104	T, GM, GTC	NR	Best performance (1 pt): sens > 90%, PPV > 50%	No	No	Heart rate changes found in 50/104 seizures. Eventual detection algorithm tested on 3 patients, varying results.
Osorio [29]	Heart rate	ECG	81 (81)	241	NR	303	1: Sens: 98%, FA: 9.5/h2: Sens: 86%, FA: 1.1/h	Yes	No	Results of the most (1) and least (2) sensitive settings are reported.
Kalitzin et al., 2012 [18]	Video	Infrared camera in room	50 (50)	72	C, GTC	NR	Sens: 95%FA: <1/24h	No	No	FA rate based on 37 h of recording without seizures. Sensitivity based on detection of seizure fragments. Nocturnal detection.
de Bruijne et al. [35]	Sound	Microphone	17 (17)	95	NR	NR	Sens: 95–98%PPV: 2–40%	No	No	Classification of seizure-related sounds and simulated nonseizure sounds. Varying performance based on sound type.

Mvt = movement; Snd = sound; ACM = accelerometry; EMG = electromyography; ECG = electrocardiography; EDA = electrodermal activity; GTC = generalized tonic–clonic; TC = tonic–clonic; C = clonic; GM = generalized myoclonic; T = tonic; CP = complex partial; HM = hypermotor; sens = sensitivity; spec = specificity; FA = false alarm rate; PPV = positive predictive value; h = hour; s = second; NR = not reported.

[a] Reported outcome measures are over all seizure types, unless specified otherwise.

[b] Intention to diagnose: "Yes" if performance of the system was based on all patients included in the study, not on a selection of patients.

Recent encouraging developments include new algorithms and existing devices. We performed a systematic literature search to critically review the performance of non-EEG, automatic systems for in-home seizure detection.

2.3.1. Methods — search strategy and selection of articles

PubMed and EMBASE databases were searched for human studies published in English up to November 2014, using variations and synonyms of search terms "*seizure prediction*" OR "*seizure detection*" *OR* "*seizures*" *AND* "*alarm*". Articles were screened for relevance to automatic, ambulatory, real-time seizure detection, based on title and abstract. If an algorithm was described in one paper and validated in another, only the validation study was included. Articles on the detection of neonatal seizures were excluded as this is a very specific area of expertise. Studies reporting results for fewer than 5 subjects were also excluded (Fig. 1). References and related articles of relevant studies were screened to check for articles not included in the search strategy.

2.3.2. Results

This search strategy yielded 723 articles of which 15 studies fulfilled the inclusion and exclusion criteria. Screening of related articles and references led to 2 additional relevant studies, making a total of 17 included studies (Fig. 1). A summary of the studies is presented in Table 1.

2.3.3. Sensor types

None of the studies evaluated EEG-based automatic seizure detection devices intended for in-home use. All studies used extracerebral signals including accelerometry ($n = 7$), mattress-based sensor ($n = 4$), heart rate ($n = 2$), sound ($n = 1$) or video ($n = 1$), a combination of accelerometry and EDA ($n = 1$), and EMG ($n = 1$) (Table 1) [12–17,19–25,32,33].

2.3.4. Study quality

All studies were performed in a hospital setting. Seven studies reported the recording time, which varied between several hours to 3 days per patient. No long-term data were reported.

2.3.5. Detection of generalized tonic–clonic seizures

Most studies (10 out of 17) focused on the detection of GTCSs. Five of these studies reported performance measures based on algorithms developed within the same study population; the other five studies reported on algorithm validation in a different population. In the former studies, sensitivity ranged from 88% to 100%, with false alarm rates between 0.1 and 1/24 h [18,22,24,26,31]. Performance was probably overestimated because of 'overfitting' of the algorithm for one specific dataset. One study focused on the detection of nocturnal seizures using automated video analysis of 72 seizures in 51 patients (sensitivity: 95%, false alarm rate: <1/24 h) [29]. Results of the other studies were based on small numbers of seizures (8–22) and included both daytime and nighttime data.

2.3.6. Detection of hypermotor seizures

Two studies evaluated the nocturnal detection of hypermotor seizures, both in the same subset of 7 subjects (who had 51 hypermotor seizures), using different methods. The first method evaluated the performance of an unsupervised detection algorithm, which does not require external validation. The second method was external validation of a previously developed detection algorithm. Sensitivities of 85% and 95.7% and positive predictive values of 58% and 46% were found [23, 25,32]. The positive predictive value (PVV) represents the proportion of correct detections, and a PPV of 60% means that 60% of the alarms generated by the device are true detections and 40% are false alarms.

2.3.7. Detection of other seizure types

Five studies also reported on the detection of other types of seizure, such as myoclonic, tonic, and complex partial seizures. Tonic and myoclonic seizures were investigated in two studies by the same research group. For both seizure types, sensitivity was 80%, with a very low positive predictive value (35% and 16%) [20,21]. The algorithms were not validated in an external population. Two other studies reported on the external validation of three mattress-based pressure sensors for the detection of GTCSs, myoclonic, tonic, and complex partial seizures. The overall performance of the sensors was poor, with an overall sensitivity of 2.2%, 4.3%, and 29.5% for the three sensors. Sensitivity per seizure type was only reported for GTCSs (sensitivities: 0%, 11.1%, and 84.6%) [28,30]. One study investigated the performance of heart rate-derived algorithms for the detection of tonic, myoclonic, and GTC seizures in three patients, with very discrepant results (sensitivity: up to 90%, PPV: generally below 50%).

2.3.8. Available devices

In five studies, the performance of existing devices was prospectively evaluated in an epilepsy monitoring unit. All studies focused on the detection of GTCSs. The ST2® (MedPage, Corby, UK), a mattress sensor for detecting nocturnal GTCSs, provided poor results [30]. This device is currently no longer available. The MP5® (MedPage, Corby, UK) mattress detector was evaluated in two studies [27,30] and had a detection rate for nocturnal GTCSs of 62.5% (5/8 seizures) with a false alarm rate of 4/24 h in one study and a detection rate of 11% (1/9 seizures, false alarm rate not reported) in the other. Why these results are so different is unclear. The Emfit® (Emfit Ltd., Vaajakoski, Finland) was evaluated in 51 adults [34] and in 45 children [28]. In adults, 18 seizures were recorded: the sensitivity of detection was 89% with a PPV of 43%. Eight nocturnal GTCSs were recorded (sensitivity: 100% and no false alarms). In children, 13 nocturnal GTCSs were recorded with a sensitivity of 84.6%; the number of false alarms was not reported. The Epi-care Free® (Danish Care Technology, Sorø, Denmark) uses accelerometry data to detect GTCSs. This device was evaluated in 73 patients in an epilepsy monitoring unit; 35 out of 39 GTCSs were detected (sensitivity: 89.7%) with a false alarm rate of 0.2/24 h [25].

2.3.9. Conclusion on clinical evaluation of available systems

The sensitivity (11% to 100%) and false alarm rates (0.2–4/24 h) of various devices for detecting GTCSs vary considerably, and for other seizure types, either detection rates are low or devices produce many false alarms. Of the four devices for the detection of GTCSs externally validated in prospective studies in a clinical setting, two had a sensitivity higher than 80% and acceptable false alarm rates: a mattress-based device for the detection of nocturnal GTCSs (sensitivity: 84.6% in children, false alarm rate not reported and sensitivity: 100% in adults, no false alarms) and a wrist-based detector of GTCSs, which can be used 24 h a day (sensitivity: 89.7%; false alarm rate: 0.2/24 h). Both devices were tested in children and adults. However, it should be mentioned that there were relatively few seizures in the validation population during the 3-day validation period. These devices need to be evaluated in a home-care setting for a longer time in order to assess the true value of these systems.

3. Benefits of seizure detection — safety

In this section, we discuss the potential benefits of automatic seizure detection in terms of safety issues, including SUDEP, status epilepticus, and physical injury due to seizures.

3.1. SUDEP

Sudden unexpected death in epilepsy is defined as "*Sudden, unexpected, witnessed or unwitnessed, nontraumatic and nondrowning death, occurring in benign circumstances, in an individual with epilepsy, with or without evidence for a seizure and excluding documented status epilepticus (seizure duration ≥30 min or seizures without recovery in between), in which postmortem examination does not reveal a cause of death*" [36].

Though SUDEP is relatively rare, with an incidence of 1.16/1000 person-years (meta-analysis by Thurman et al. [37]), the lifetime risk of SUDEP in people with epilepsy is between 8% and 12% [37], making it a very relevant issue in epilepsy care. The incidence of SUDEP is higher in specific groups, such as people with refractory epilepsy (3.8–9.3/1000 person-years, median: 6.2/1000 person-years) [38] and people with learning disabilities (3.4–3.6/1000) [39].

3.1.1. Can detection of seizures help prevent SUDEP?

A meta-analysis of four case–control studies showed that a history of GTCSs is the main risk factor for SUDEP (OR: 5, 95% CI: 3–8.6, in subjects with 1–2 GTCSs/year and OR: 15.6, 95% CI: 10.1–24.2, in subjects with >2 GTCSs/year) [40]. Other risk factors include male sex, symptomatic epilepsy, and duration of epilepsy. Sudden unexpected death in epilepsy is often a sleep-related (58%) and unwitnessed (86%) event [41]. In one case–control study, nocturnal seizure was reported as independent risk factor (OR: 2.6, 95%: CI 1.3–5.0) and nocturnal supervision a protective factor (OR: 0.4, 95% CI: 0.2 to 0.8) for SUDEP [41,42]. Unfortunately, these factors have not been evaluated in other studies. In a cohort of children with learning disability attending a school with very close supervision, all 14 SUDEP cases occurred during holidays when the children were under less intensive supervision. As most cases are unwitnessed, supervision would seem to have a protective effect [37].

It is not known why SUDEP is more often unwitnessed than witnessed, but it has been proposed that timely stimulation (e.g., through shaking) of a subject with cardiorespiratory depression following a seizure might help arouse the subject and restore cardiorespiratory function [43,44]. The scientific and clinical epilepsy community recommends nocturnal supervision in high-risk subjects, although it is debated how this should be done [43,44]. Continuous supervision for a low-frequency event during the night is not realistic in most care settings, either at home or in institutions. Better monitoring of especially nocturnal seizures with automatic seizure detectors would give caregivers the opportunity to intervene, and this might be a way to prevent SUDEP. As long as SUDEP pathophysiology remains elusive, it is unlikely that specific SUDEP prevention devices will be developed soon. However, as SUDEP is closely related to GTCSs, a sensitive ambulatory device detecting GTCSs is a realistic first step towards SUDEP prevention. It will be a challenge to obtain scientific proof for this first step as this will require large, long-term follow-up studies.

3.2. Other aspects of safety

Status epilepticus (SE) and injuries related to seizures are a major risk to patients. The annual incidence of SE is about 10–20/100,000 [45] of which about a third occur in people with a history of epilepsy [46,47]. The earlier an intervention can take place to stop tonic–clonic SE, the better the prognosis is in terms of mortality, cognition, and functional outcome [48]. More accurate and earlier detection of seizures will give caregivers the opportunity to administer emergency medication and, thus, reduce treatment delay.

The absolute risk of seizure-related injuries ranges from 12% to 35% in various populations (general population, institutionalized, children, and subjects with and without learning disability) [49]. Most injuries are due to falls, burns, traffic accidents, and swimming-related incidents. Nocturnal seizures may cause the subject to fall out of bed, aspirate, or wander (postictal confusional wandering). The incidence of these complications is not clear from the literature. Apart from postictal wandering, it is unlikely that automatic seizure detection will help prevent injuries as these events mostly occur at seizure onset.

4. Benefits of seizure detection — caregiver perspective

In the following sections, we discuss the benefits of seizure detection regarding other stakeholders in epilepsy care, i.e., caregivers at home and professional caregivers, clinicians, and researchers.

4.1. Caregivers at home

The caregivers of people with epilepsy feel that they should be there when a potentially harmful seizure occurs. This psychological burden affects the health-related QoL of caregivers [50], but the extent to which their health-related QoL is affected depends on their personal coping style. A passive response style towards negative events is associated with a worse health-related QoL [51]. The parents of children with epilepsy often sleep in the same bedroom or next to their child in the hope of detecting or preventing nocturnal seizures. These parents are more fatigued and experience more disturbed sleep than the parents of healthy children [52]. Moreover, these sleeping arrangements can have detrimental effects on the love life of the parents and negatively influence the child's development.

Installation of a video observation system was found to improve parent-reported quality of family life [53]. However, this requires continuous surveillance of the video images, which is still a burden. Reliable nocturnal seizure monitoring with a seizure detector may help caregivers cope better with the uncertainty of epilepsy and improve their QoL.

4.2. Professional care setting

A significant number of people with refractory epilepsy also have intellectual disability that necessitates their receiving specialized care. In our experience, although these institutions aim to provide the same level of care and safety as at home, they are dependent on the availability of nursing staff for nocturnal supervision. Sudden unexpected death in epilepsy in an institutional setting may cause a lot of commotion and media attention. This is understandable, but expectations of the care provided cannot always be met. Currently, no guidelines indicate how these high-risk institutionalized individuals should be monitored. Camera supervision alone is not likely to solve the problem, because continuous visual inspection of multiple individuals requires numerous personnel and is tedious, so that there is a high chance of an event being missed. Optimal automatic seizure detection would enable better and more efficient monitoring of high-risk individuals, both with regard to manpower and costs. Care professionals can devote their time to caregiving instead of monitoring videoscreens.

5. Benefits of seizure detection — clinical perspective

In-home seizure monitoring may provide clinicians with valuable information. Self-reported seizure frequencies are notoriously unreliable, especially in the case of nocturnal seizures, as has been shown by nocturnal video-EEG and intracranial recordings [14,54]. Long-term monitoring in a video-EEG monitoring unit is costly, and the change of environment often influences seizure frequency. More possibilities for in-home monitoring and telecare could reduce the number of hospital visits [55].

It is disputable whether accurate knowledge of seizure frequency will lead to more effective treatment strategies — people with refractory epilepsy have often unsuccessfully tried a number of treatment options. However, in early stages of the disease, more accurate information may improve diagnosis and treatment, which could influence the disease course. This is especially the case with epileptic encephalopathy [56].

6. Benefits of seizure detection — research perspective

In-home seizure detection could improve the quality of clinical research. In clinical trials, the effects of new drugs or other interventions should ideally be monitored with an objective measure. There is relatively little research-based knowledge of seizure frequency, seizure semiology, and response to treatment of people with refractory epilepsy and learning disabilities. The majority of especially nocturnal seizures go unnoticed in people with learning disabilities living in a residential care setting [57]. A reliable seizure detection system may help avoid undertreatment of these subjects.

7. Disadvantages of seizure detection

There are also disadvantages to seizure detection in epilepsy care. Primarily, an automatic device is never foolproof and could give caregivers a false sense of security. Moreover, caregivers need to be able to cope with the system and use it correctly. They should also appreciate the limits and uses of a detection system — a seizure detector is not necessarily a vital sign monitor. Thus, clear information about what a detector can and cannot do is needed to ensure the proper use and acceptance of a device.

Another aspect of automatic detection is the risk of "alarm fatigue". Seizures occur in varying frequencies and not all seizures require care. Every device will also occasionally produce false alarms. If only a small proportion of alarms is relevant to the caregiver, then the caregiver will stop responding to alarms as these alarms tend to be "nothing" in most occasions [58]. A device that leads to such alarm fatigue defeats its purpose.

Also, the availability of devices might stimulate unnecessary monitoring by overprotective caregivers. A child might (un)consciously trigger the alarm to attract the attention of the caregiver. Furthermore, monitoring might reveal a higher seizure frequency than expected or show that treatment does not make any difference. This knowledge can be hard for caregivers to cope with if treatment options are limited. An alarm system gives carers the responsibility to respond to an alarm and to provide proper ictal and postictal care, and carers will feel guilty if they ignore a 'real' alarm.

Privacy is also an issue with video monitoring systems, especially in children and people with learning disabilities. This is particularly important in the context of sexual behavior (for example, masturbation might trigger a false alarm).

8. Economic evaluation of seizure detection

Automatic seizure detection should be affordable, but detectors for continuous monitoring are expensive. However, as epilepsy detectors might reduce costs, by reducing the need for long-term in-hospital diagnostic monitoring, reducing the need for 24-hour care, and reducing hospital admissions for SE, health insurance companies might be persuaded to cover the costs of these systems. An economic cost–benefit analysis will be complex, as potential benefits in health-care utilization will probably be outweighed by benefits to the quality of life of caregivers.

9. Seizure monitoring in a broader context

Personalized health monitoring in a home setting is an emerging trend in modern health care. Examples are monitoring of oxygen saturation in people with chronic obstructive pulmonary disease (COPD), glucose monitoring in people with diabetes, and weight monitoring in people with heart failure. In these cases, the goal is to obtain ambulatory data so that health-care professionals can decide if and what further action/treatment is necessary. The first large-scale trial of this type of monitoring in people with COPD, diabetes, and heart failure showed promising overall results at 12 months, with a reduced likelihood of hospital admission (OR: 0.82, 95% CI: 0.70–0.97) and death (OR: 0.54 95% CI: 0.37–0.75) [52]. In pediatric asthma care, electronic evaluation of adherence to inhaled steroids has led to valuable insights into reasons for nonadherence, which has enabled pediatricians to provide better information on medication use [59]. Whether seizure monitoring will provide comparable benefits is difficult to say, but studies have shown that disease monitoring in the home setting provides valuable information not obtained with monitoring in the hospital setting.

Other developments might find their parallel in epilepsy monitoring. For example, in COPD, care algorithms are being developed to predict exacerbation based on several clinical parameters [60]. In the care for elderly people, camera or radar technology could be used to monitor falls and nocturnal wandering. To date, there have been no studies of the safety and influence on QoL and perceived privacy of the habitants of this approach. Elderly people have expressed their concern about privacy issues. However, most people are willing to give up a certain amount of privacy to maintain autonomy and to be able to continue living in their own home as opposed to moving to a residential care facility [61]. More needs to be learned about the attitude of people with epilepsy regarding autonomy and privacy before online monitoring can be considered a viable option.

10. Future perspectives

In our opinion, automatic seizure detectors can be used in three main settings: (1) home-based monitoring with alarm, (2) monitoring with alarm in institutionalized subjects, and (3) in-hospital or home-based recording with a diagnostic or clinical research purpose.

When considering home-based, monitoring with alarm, an optimal seizure detector will detect only seizures that require an intervention. Alarms are especially relevant during the night, so caregivers can sleep in separate bedrooms. The tolerance for false alarms during the night will be low, as they will unnecessarily wake a caregiver. A future system needs to be simple to use in a home setting and not only function in a controlled research environment with highly trained caregivers.

In an institutional setting, detection of seizures that need intervention is also needed, but the definition of 'a seizure needing intervention' is more focused on situations where the subject requires medical attention. Especially during the night, one caregiver will attend to multiple subjects who may not reside in the same location. This means that the false alarm rate needs to be relatively low, as the caregiver could have to cope with multiple alarms depending on the number of residents. On the other hand, online camera images can help make false alarms more manageable by giving the opportunity of a fast visual check of the state of the subject. The availability of camera images at a central location is an extra requirement for a useful automatic seizure detector in this setting. Especially in institutions with many residents, monitoring can become a high financial burden, so systems need to be as cheap as possible. In the MORTEMUS study of registered cases of SUDEP, in all cases, a time window of at least 3 min after the end of the seizure and before the start of fatal respiratory and cardiac disruptions was observed [62]. This indicates that there is time for a professional caregiver to assess the urgency of an automatically detected seizure through camera images and, subsequently, to decide if physical attendance is needed. In this scenario, caregivers should be able to provide proper measures of resuscitation [44]. When the aim of a system is purely SUDEP prevention, vital sign monitoring instead of seizure detection could also be an option. Reliable detection of prolonged cardiorespiratory dysfunction is a possibility; however, this is only effective when very immediate care can be provided.

For seizure detection with a diagnostic or clinical research purpose, a seizure detector needs to meet different requirements. Many different seizure types need to be detected and classified. As the purpose here is diagnostic, no alarm is necessary and false detections will not have direct practical care consequences.

Based on the known setting and most prevalent seizure types, it is probably feasible to optimize seizure detection by adapting the detection algorithms to the situation. Techniques such as active learning [63] can further improve a personalized system over time by incorporating feedback on the correctness of detections. If necessary, this can be done in a professional care setting where each detection can be evaluated.

10. Conclusion

Not all types of seizures can be successfully detected with existing seizure detection algorithms and devices. Automatic detection of especially nocturnal GTCSs seems feasible, though results vary. More studies in the home setting and involving more subjects with long-term recordings are necessary because the general behavior and semiology of seizures can be different, influencing algorithm or device performance. Studies tend to be overoptimistic when they report on the performance of GTCS detection based on the same data used to develop the algorithm [18,22,26,31]. These algorithms need to be tested in an independent population before we know their true value in practice.

Further development of seizure detectors could help improve epilepsy care. Automatic seizure detection could play an important role in SUDEP prevention, facilitating early intervention in SE, improving the QoL of caregivers, monitoring drug therapy, and improving long-term diagnostic investigations and the quality of clinical research. Available detectors and algorithms are inadequate for detecting other seizure types, such as hypermotor seizures or generalized tonic seizures.

Future research should involve long-term data collection in a home-care setting. Detector performance should be expressed as the ‘number needed to warn’, representing the number of alarms generated to detect one clinically relevant seizure. This number will differ depending on seizure frequency and can reflect the usefulness of monitoring in specific patient groups. The benefits and risks of seizure detection should be monitored, taking into account the QoL of caregivers, the comfort and privacy of subjects, and the impact of false alarms. Other factors to consider include the increased number of interventions because of SE and the effects of more accurate knowledge of seizure frequency on compliance and treatment decisions. Such an integrative approach could provide insight into the societal benefit of seizure detection in epilepsy care.

Acknowledgments

This study was funded by ZonMW, grant number 300040003 and by the Dutch Epilepsy Foundation. The funders had no role in study design, data collection and analysis, decision to publish, or preparation of the manuscript.

Conflict of interest statement

None of the authors has a personal, commercial interest in automatic seizure detection.

RDT receives research support from the Dutch Epilepsy Foundations (project number 15-10), NUTS Ohra Fund, Medtronic, Christelijke Vereniging voor de Verpleging van Lijders aan Epilepsie and AC Thomson Foundation. RDT has received fees for lectures from Medtronic, UCB Pharma, and GSK.

References

[1] Coatsworth JJ. NINDS monograph, 12. Bethesda, MD: US Department of Health and Education; 1971.
[2] Löscher W, Schmidt D. Modern antiepileptic drug development has failed to deliver: ways out of the current dilemma. Epilepsia 2011;52(4):657–78.
[3] Hauser W, Allen F, Annegers J, Elveback LR. Mortality in patients with epilepsy. Epilepsia 1980;21(4):399–412.
[4] Neligan A, Bell GS, Shorvon SD, Sander JW. Temporal trends in the mortality of people with epilepsy: a review. Epilepsia 2010;51(11):2241–6.
[5] Shorvon S. The epidemiology and treatment of chronic and refractory epilepsy. Epilepsia 1996;37(Suppl. 2):S1–3.
[6] Jacoby A, Snape D, Baker GA. Determinants of quality of life in people with epilepsy. Neurol Clin 2009;27(4):843–63.
[7] Mormann F, Andrzejak RG, Elger CE, Lehnertz K. Seizure prediction: the long and winding road. Brain 2007;130:314–33.
[8] Cook MJ, O'Brien TJ, Berkovic SF, Murphy M, Morokoff A, Fabinyi G, et al. Himes D prediction of seizure likelihood with a long-term, implanted seizure advisory system in patients with drug-resistant epilepsy: a first-in-man study. Lancet Neurol 2013; 12(6):563–71.
[9] Pauri F, Pierelli F, Chatrian GE, Erdly WW. Long-term EEG-video-audio monitoring: computer detection of focal EEG seizure patterns. Electroencephalogr Clin Neurophysiol 1992;82:1–9.
[10] Hopfengärtner R, Kasper BS, Graf W, Gollwitzer S, Kreiselmeyer G, Stefan H, Hamer H. Automatic seizure detection in long-term scalp EEG using an adaptive thresholding technique: a validation study for clinical routine. Clin Neurophysiol 2014;125(7):1346–52.
[11] Ramgopal S, Thome-Souza S, Jackson M, Kadish NE, Fernández IS, Klehm J, et al. Seizure detection, seizure prediction, and closed-loop warning systems in epilepsy. Epilepsy Behav 2009;37:291–307.
[12] Duun-Henriksen J, Madsen RE, Remvig LS, Thomsen CE, Sorensen HB, Kjaer TW. Automatic detection of childhood absence epilepsy seizures: toward a monitoring device. Pediatr Neurol 2012;46(5):287–92.
[13] Donos C, Dümpelmann M, Schulze-Bonhage A. Early seizure detection algorithm based on intracranial EEG and random forest classification. Int J Neural Syst 2015; 25(5), 1550023. http://dx.doi.org/10.1142/S0129065715500239.
[14] Bhardwaj A, Tiwari A, Krishna R, Varma V. A novel genetic programming approach for epileptic seizure detection. Comput Methods Prog Biomed 2016; 124:2–18.
[15] Logesparan L, Casson AJ, Rodriguez-Villegas E. Optimal features for online seizure detection. Med Biol Eng Comput 2012;50(7):659–69.
[16] Juhl CB, Højlund K, Elsborg R, Poulsen MK, Selmar PE, Holst JJ, et al. Automated detection of hypoglycemia-induced EEG changes recorded by subcutaneous electrodes in subjects with type 1 diabetes — the brain as a biosensor. Diabetes Res Clin Pract 2010;88(1):22–8.
[17] Osorio I, Schachter S. Extracerebral detection of seizures: a new era in epileptology? Epilepsy Behav 2011;22:S82–7.
[18] Kalitzin S, Petkov G, Velis D, Vledder B, Lopes da Silva F. Automatic segmentation of episodes containing epileptic clonic seizures in video sequences. IEEE Trans Biomed Eng 2012;59(12):3379–85.
[19] Van de Vel A, Cuppens K, Bonroy B, Milosevic M, Jansen K, Van Huffel S, et al. Ceulemans B non-EEG seizure-detection systems and potential SUDEP prevention: state of the art. Seizure 2013;22(5):345–55.
[20] Nijsen TME, Aarts RM, Arends JBAM, Cluitmans PJM. Automated detection of tonic seizures using 3-D accelerometry. IFMBE Proc 2008;22:188–91.
[21] Nijsen TM, Aarts RM, Cluitmans PJ, Griep PA. Time–frequency analysis of accelerometry data for detection of myoclonic seizures. IEEE Trans Inf Technol Biomed 2010;14(5):1197–203.
[22] Kramer U, Kipervasser S, Shlitner A, Kuzniecky R. A novel portable seizure detection alarm system: preliminary results. J Clin Neurophysiol 2011;28(1):36–8.
[23] Van de Vel A, Cuppens K, Bonroy B, Milosevic M, Van Huffel S, Vanrumste B, et al. Long-term home monitoring of hypermotor seizures by patient-worn accelerometers. Epilepsy Behav 2013;26(1):118–25.
[24] Lockman J, Fisher RS, Olson DM. Detection of seizure-like movements using a wrist accelerometer. Epilepsy Behav 2011;20(4):638–41.
[25] Beniczky S, Polster T, Kjaer TW, Hjalgrim H. Detection of generalized tonic–clonic seizures by a wireless wrist accelerometer: a prospective, multicenter study. Epilepsia 2013;54(4):e58–61.
[26] Poh MZ, Loddenkemper T, Reinsberger C, Swenson NC, Goyal S, Sabtala MC, et al. Convulsive seizure detection using a wrist-worn electrodermal activity and accelerometry biosensor. Epilepsia 2012;53(5):e93–7.
[27] Carlson C, Arnedo V, Cahill M, Devinsky O. Detecting nocturnal convulsions: efficacy of the MP5 monitor. Seizure 2009;18(3):225–7.
[28] Van Poppel K, Fulton SP, McGregor A, Ellis M, Patters A, Wheless J. Prospective study of the emfit movement monitor. J Child Neurol 2013;28(11):1434–6.
[29] Osorio I. Automated seizure detection using EKG. Int J Neural Syst 2014;24(02).
[30] Fulton S, Van Poppel K, McGregor A, Ellis M, Patters A, Wheless J. Prospective study of 2 bed alarms for detection of nocturnal seizures. J Child Neurol 2013;28(11): 1430–3.
[31] Conradsen I, Beniczky S, Hoppe K, Wolf P, Sorensen HB. Automated algorithm for generalized tonic–clonic epileptic seizure onset detection based on semg zero-crossing rate. IEEE Trans Biomed Eng 2012;59(2):579–85.
[32] Luca S, Karsmakers P, Cuppens K, Croonenborghs T, Van de Vel A, Ceulemans B, et al. Detecting rare events using extreme value statistics applied to epileptic convulsions in children. Artif Intell Med 2014;60(2):89–96.
[33] van Elmpt WJ, Nijsen TM, Griep PA, Arends JB. A model of heart rate changes to detect seizures in severe epilepsy. Seizure 2006;15(6):366–75.
[34] Narechania AP, Garic II, Sen-Gupta I, Macken M, Gerard EE, Schuele SU. Assessment of a quasi-piezoelectric mattress monitor as a detection system for generalized convulsions. Epilepsy Behav 2013;28:172–6.
[35] de Bruijne GR, Sommen PCW, Aarts RM. Detection of epileptic seizures through audio classification. 4th European Conference of the International Federation for Medical and Biological Engineering. Berlin Heidelberg: Springer; 2009. p. 1450–4.
[36] Nashef L, So EL, Ryvlin P, Tomson T. Unifying the definitions of sudden unexpected death in epilepsy. Epilepsia 2012;53(2):227–33.
[37] Thurman DJ, Hesdorffer DC, French JA. Sudden unexpected death in epilepsy: assessing the public health burden. Epilepsia 2014;55(10):1479–85.
[38] Tomson T, Nashef L, Ryvlin P. Sudden unexpected death in epilepsy: current knowledge and future directions. Lancet Neurol 2008;7(11):1021–31.
[39] Téllez-Zenteno JF, Hernández Ronquillo L, Wiebe S. Sudden unexpected death in epilepsy: evidence-based analysis of incidence and risk factors. Epilepsy Res 2005; 65:101–15.
[40] Hesdorffer DC, Tomson T, Benn E, Sander JW, Nilsson L, Langan Y, et al. Do antiepileptic drugs or generalized tonic–clonic seizure frequency increase SUDEP risk? A combined analysis. Epilepsia 2012;53(2):249–52.

[41] Lamberts RJ, Thijs RD, Laffan A, Langan Y, Sander JW. Sudden unexpected death in epilepsy: people with nocturnal seizures may be at highest risk. Epilepsia 2012;53(2):253–7.
[42] Langan Y, Nashef L, Sander JW. Case–control study of SUDEP. Neurology 2005;64(7):1131–3.
[43] Shorvon S, Tomson T. Sudden unexpected death in epilepsy. Lancet 2011;378:2028–38.
[44] Ryvlin P, Nashef L, Tomson T. Prevention of sudden unexpected death in epilepsy: a realistic goal? Epilepsia 2013;54(S2):23–8.
[45] Rosenow F, Hamer HM, Knake S. Epidemiology of convulsive and nonconvulsive status epilepticus. Epilepsia 2007;48(S8):82–4.
[46] Knake S, Rosenow F, Vescove M, Oertel WH, Mueller H-H, Wirbatz A, Katsarou N, Hamer HM. Incidence of status epilepticus in adults in Germany: a prospective, population-based study. Epilepsia 2001;42(6):714–8.
[47] DeLorenzo RJ, Hauser WA, Towne AR, Boggs JG, Pellock JM, Penberthy L, et al. A prospective, population-based epidemiologic study of status epilepticus in Richmond, Virginia. Neurology 1996;46:1029–35.
[48] Neligan A, Shorvon SD. Prognostic factors, morbidity and mortality in tonic–clonic status epilepticus: a review. Epilepsy Res 2010;93:1–10.
[49] Nguyen R, Téllez Zenteno JF. Injuries in epilepsy: a review of its prevalence, risk factors, type of injuries and prevention. Neurol Int 2009;16(1):e20.
[50] Van Andel J, Zijlmans M, Fischer K. Leijten FSS quality of life of caregivers of patients with intractable epilepsy. Epilepsia 2009;50(5):1294–6.
[51] Van Andel J, Westerhuis W, Zijlmans M, Fischer K, Leijten FSS. Coping style and health-realted quality of life in caregivers of people with epilepsy. J Neurol 2011;258:1788–94.
[52] Larson AM, Ryther RCC, Jennesson M, Geffrey AL, Bruno PL, Anagnos CJ, et al. Impact of pediatric epilepsy on sleep patterns and behaviors in children and parents. Epilepsia 2012;53(7):1162–9.
[53] Johansen JR, Lindahl G, Sandstedt P. Home-video observation of seizures in children with epilepsy-impact on quality of family life. Seizure 1999;8:356–7.
[54] Eisenman LN, Attarian H, Fessler AJ, Vahle VJ, Gilliam F. Self-reported seizure frequency and time to first event in the seizure monitoring unit. Epilepsia 2005;46(5):664–8.
[55] Elger CE, Burr W. Advances in telecommunications concerning epilepsy. Epilepsia 2000;41(Suppl. 5):S9–S12.
[56] Vigevano F, Arzimanoglou A, Plouin P, Specchio N. Therapeutic approach to epileptic encephalopathies. Epilepsia 2013;54(s8):45–50.
[57] Nijsen TME, Arends JBAM, Griep PAM, Cluitmans PJM. The potential value of three-dimensional accelerometry for detection of motor seizures in severe epilepsy. Epilepsy Behav 2005;7(1):74–84.
[58] Edworthy J. Medical audible alarms: a review. J Am Med Inform Assoc 2013;20(3):584–9.
[59] Morton RW, Everard ML, Elphick HE. Adherence in childhood asthma: the elephant in the room. Arch Dis Child 2014;99(10):949–53.
[60] Pinnock H, Hanley J, McCloughan L, Todd A, Krishan A, Lewis S, et al. Effectiveness of telemonitoring integrated into existing clinical services on hospital admission for exacerbation of chronic obstructive pulmonary disease: researcher blind, multicentre, randomised controlled trial. BMJ 2013;17:347.
[61] Townsend D, Knoefel F, Goubran R. Privacy versus autonomy: a tradeoff model for smart home monitoring technologies. Conf Proc IEEE Eng Med Biol Soc 2011:4749–52.
[62] Ryvlin P, Nashef L, Lhatoo SD, Bateman LM, Bird J, Bleasel A, et al. Incidence and mechanisms of cardiorespiratory arrests in epilepsy monitoring units (MORTEMUS): a retrospective study. Lancet Neurol 2013;12:966–77.
[63] Cohn D, Atlas L, Ladner R. Improving generalization with active learning. Mach Learn 1994;15(2):201–21.